5.6 Proving Triangle Congruence by ASA and AAS

Name _____ Period ____ Date _____

Geometry 5.6 · Student sheet

I can…

- use ASA to decide whether triangles are congruent.
- use AAS to decide whether triangles are congruent.
- tell which combinations of parts do not prove congruence.

Vocabulary · fill in the missing word

Included side — the side between two angles of a _____

ASA Congruence Theorem — two angles and the _____ side of one triangle congruent to two angles and the included side of another

Part 1 · Discovery: two angles and a side

Draw a 6 cm segment with a 40° angle at one end and a 70° angle at the other. Then draw any triangle with angles 40°, 70°, 70°.

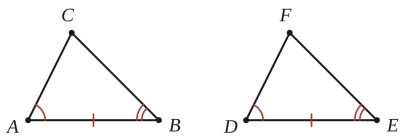
Triangle	The other two sides	Matched my neighbor's exactly?
40°, 6 cm, 70°		
40°, 70°, 70° only		

Part 2 · Notes and examples (fill in each note, then work the example)

ASA Congruence Theorem: if two angles and the _____ side of one triangle are congruent to two angles and the included side of a second triangle, then the triangles are congruent.

Example 1

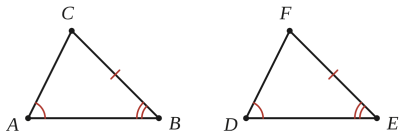
$\angle A \cong \angle D$, $\overline{AB} \cong \overline{DE}$, and $\angle B \cong \angle E$. Are $\triangle ABC$ and $\triangle DEF$ congruent? Which theorem?



AAS Congruence Theorem: if two angles and a _____ side of one triangle are congruent to two angles and the corresponding non-included side of a second triangle, then the triangles are congruent.

Example 2

$\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\overline{BC} \cong \overline{EF}$. Are $\triangle ABC$ and $\triangle DEF$ congruent? Which theorem?



Example 3

Given: $\angle B \cong \angle D$ and $\overline{AC}$ bisects $\angle BAD$. Prove: $\triangle ABC \cong \triangle ADC$.

Not tests: AAA and SSA do _____ prove triangles congruent.

Example 4

Which theorem, if any, proves the triangles congruent?

a. all three pairs of angles congruent

b. two pairs of sides and a non-included pair of angles congruent

c. two pairs of angles and the included pair of sides congruent

Example 5

A surveyor measures the angles at two points on shore and the distance between those points. Why does that fix the triangle (and the distance across the lake)?

Part 3 · Partner practice

Partner 1

In $\triangle PQR$ and $\triangle XYZ$: $\angle P \cong \angle X$, $\angle Q \cong \angle Y$, and $\overline{PR} \cong \overline{XZ}$. Which theorem proves them congruent?

Partner 2

$\angle ABD \cong \angle CBD$ and $\angle ADB \cong \angle CDB$, and the triangles share $\overline{BD}$. Which theorem proves $\triangle ABD \cong \triangle CBD$?